

2024 AMC 10B Problem 24 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 24. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 24

Problem:

Let

$$P(m) = \frac{m}{2} + \frac{m^2}{4} + \frac{m^4}{8} + \frac{m^8}{8}.$$

How many of the values $P(2022)$, $P(2023)$, $P(2024)$, and $P(2025)$ are integers?

Answer Choices:

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

Solution:

Let $Q(m) = 8P(m) = 4m + 2m^2 + m^4 + m^8$. Because the coefficients of Q are integers, it follows that if $a \equiv b \pmod{8}$, then $Q(a) \equiv Q(b) \pmod{8}$. It suffices to show that the 8 numbers $Q(-3)$, $Q(-2)$, $Q(-1)$, $\dots$, $Q(4)$ are all divisible by 8. If m is even, then each of the

monomials of $Q(m)$ is divisible by 8. If $m = \pm 1$, then $Q(m) = \pm 4 + 4 \equiv 0 \pmod{8}$. If $m = \pm 3$, then $m^2 = 9 \equiv 1 \pmod{8}$, which implies that $m^4 \equiv 1 \pmod{8}$, and so also that $m^8 \equiv 1 \pmod{8}$. Hence $Q(\pm 3) \equiv \pm 12 + 2 + 1 + 1 \equiv 0 \pmod{8}$.

Therefore $8P(m)$ is divisible by 8 for all integers m , which implies that $P(m)$ is an integer for all m . In particular, all (E) 4 of the given values of $P(m)$ are integers.

OR

Let $Q(m)$ be defined as in the first solution, and note that $Q(m)$ is divisible by 8 if m is even. To treat odd m , write

$$\begin{aligned} Q(m) &= 8m + 4(m^2 - m) + 2m^2(m^2 - 1) + m^4(m^4 - 1) \\ &= 8m + 4m(m - 1) + 2m^2(m + 1)(m - 1) + m^4(m^2 + 1)(m + 1)(m - 1) \end{aligned}$$

and note that because $m + 1$, $m - 1$, and $m^2 + 1$ are all even, each term has at least three factors of 2. The solution concludes as above.

OR

Another way to see that $Q(m)$ is divisible by 8 when m is odd in the second solution is to apply more general facts from number theory. Fermat's Little Theorem asserts that if p is prime, then $a^p \equiv a \pmod{p}$ for all integers a . In particular, $m^2 \equiv m \pmod{2}$. Euler's Totient Theorem asserts that if $\gcd(a, q) = 1$, then $a^{\phi(q)} \equiv 1 \pmod{q}$, where $\phi(q)$ is the number of positive integers less than q that are relatively prime to q . Because $\phi(4) = 2$, it follows that $m^2 \equiv 1 \pmod{4}$ when m is odd. Also, $\phi(8) = 4$, so $m^4 \equiv 1 \pmod{8}$ if m is odd.

Note: Suppose that a necklace is to be made by placing n beads, equally spaced, on a circular ring. Each bead can be any one of m different colors. Two necklaces are regarded as being the same if they differ only by rotation. In 1892 Captain Percy Alexander MacMahon showed that the number of such necklaces is

$$\frac{1}{n} \sum_{d|n} \phi\left(\frac{n}{d}\right) m^d$$

He acknowledged its prior discovery by another soldier, Monsieur le Colonel Charles Paul Narcisse Moreau. Thus $P(m)$ counts the number of necklaces with $n = 8$ beads and therefore must be an integer. This is discussed in Concrete Mathematics by Graham, Knuth, and Patashnik.

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